

PERCENTAGES

Short Revision Notes | Apply Level — Use Knowledge in New Situations

1. Applying Profit and Loss Formulas

Operation	Formula	Example
Find Profit	$SP - CP$	$CP=500, SP=650 \rightarrow \text{Profit}=150$
Find Loss	$CP - SP$	$CP=500, SP=450 \rightarrow \text{Loss}=50$
Find SP with Profit	$CP + \text{Profit}$	$CP=500, \text{Profit}=150 \rightarrow SP=650$
Find SP with Loss	$CP - \text{Loss}$	$CP=500, \text{Loss}=50 \rightarrow SP=450$
Find CP with Profit	$SP - \text{Profit}$	$SP=650, \text{Profit}=150 \rightarrow CP=500$
Find CP with Loss	$SP + \text{Loss}$	$SP=450, \text{Loss}=50 \rightarrow CP=500$

Quick Practice:

Given	Find	Method	Answer
$CP=800, SP=900$	Profit	$900-800$	100
$CP=800, SP=700$	Loss	$800-700$	100
$CP=1200, \text{Profit}=180$	SP	$1200+180$	1380
$CP=1200, \text{Loss}=240$	SP	$1200-240$	960

2. Applying Profit Percentage

Σ Formula: $\text{Profit}\% = (\text{Profit} \div CP) \times 100\%$

Problem	Step 1: Find Profit	Step 2: Apply Formula	Answer
$CP=500, SP=600$	$600-500=100$	$100/500 \times 100$	20%
$CP=800, SP=1000$	$1000-800=200$	$200/800 \times 100$	25%
$CP=1500, SP=1800$	$1800-1500=300$	$300/1500 \times 100$	20%
$CP=250, SP=325$	$325-250=75$	$75/250 \times 100$	30%
$CP=400, SP=480$	$480-400=80$	$80/400 \times 100$	20%

Quick Application:

CP	SP	Profit	Profit %
600	720	120	20%
800	1000	200	25%
1200	1500	300	25%
200	250	50	25%

3. Applying Loss Percentage

Σ Formula: $\text{Loss\%} = (\text{Loss} \div \text{CP}) \times 100\%$

Problem	Step 1: Find Loss	Step 2: Apply Formula	Answer
CP=500, SP=450	500-450=50	50/500×100	10%
CP=800, SP=720	800-720=80	80/800×100	10%
CP=1500, SP=1200	1500-1200=300	300/1500×100	20%
CP=400, SP=300	400-300=100	100/400×100	25%
CP=1000, SP=850	1000-850=150	150/1000×100	15%

Quick Application:

CP	SP	Loss	Loss %
600	540	60	10%
800	680	120	15%
1000	800	200	20%
500	450	50	10%

4. Finding Selling Price from Profit %

Σ Formula: $SP = CP \times (100 + \text{Profit}\%) \div 100$

Problem	Step 1	Step 2	Step 3	Answer
CP=500, Profit=10%	$100+10=110$	$500 \times 110 = 55,000$	$\div 100 = 550$	Rs 550
CP=800, Profit=25%	$100+25=125$	$800 \times 125 = 100,000$	$\div 100 = 1000$	Rs 1000
CP=1200, Profit=20%	$100+20=120$	$1200 \times 120 = 144,000$	$\div 100 = 1440$	Rs 1440
CP=250, Profit=30%	$100+30=130$	$250 \times 130 = 32,500$	$\div 100 = 325$	Rs 325
CP=1500, Profit=15%	$100+15=115$	$1500 \times 115 = 172,500$	$\div 100 = 1725$	Rs 1725

Quick Application:

CP	Profit %	SP
600	20%	720
800	25%	1000
400	10%	440
1000	30%	1300

5. Finding Selling Price from Loss %

Σ Formula: $SP = CP \times (100 - \text{Loss}\%) \div 100$

Problem	Step 1	Step 2	Step 3	Answer
CP=500, Loss=10%	$100-10=90$	$500 \times 90 = 45,000$	$\div 100 = 450$	Rs 450
CP=800, Loss=15%	$100-15=85$	$800 \times 85 = 68,000$	$\div 100 = 680$	Rs 680
CP=1200, Loss=20%	$100-20=80$	$1200 \times 80 = 96,000$	$\div 100 = 960$	Rs 960
CP=250, Loss=5%	$100-5=95$	$250 \times 95 = 23,750$	$\div 100 = 237.50$	Rs 237.50
CP=1000, Loss=25%	$100-25=75$	$1000 \times 75 = 75,000$	$\div 100 = 750$	Rs 750

Quick Application:

CP	Loss %	SP
600	10%	540
800	20%	640
500	5%	475
1000	15%	850

6. Finding Cost Price from Profit %

Σ Formula: $CP = SP \times 100 \div (100 + \text{Profit}\%)$

Problem	Step 1	Step 2	Step 3	Answer
SP=550, Profit=10%	$100+10=110$	$550 \div 110=5$	$5 \times 100=500$	Rs 500
SP=1000, Profit=25%	$100+25=125$	$1000 \div 125=8$	$8 \times 100=800$	Rs 800
SP=1440, Profit=20%	$100+20=120$	$1440 \div 120=12$	$12 \times 100=1200$	Rs 1200
SP=325, Profit=30%	$100+30=130$	$325 \div 130=2.5$	$2.5 \times 100=250$	Rs 250
SP=1725, Profit=15%	$100+15=115$	$1725 \div 115=15$	$15 \times 100=1500$	Rs 1500

Quick Application:

SP	Profit %	CP
720	20%	600
1000	25%	800
440	10%	400
1300	30%	1000

7. Finding Cost Price from Loss %

Σ Formula: $CP = SP \times 100 \div (100 - \text{Loss}\%)$

Problem	Step 1	Step 2	Step 3	Answer
SP=450, Loss=10%	$100-10=90$	$450 \div 90=5$	$5 \times 100=500$	Rs 500
SP=680, Loss=15%	$100-15=85$	$680 \div 85=8$	$8 \times 100=800$	Rs 800
SP=960, Loss=20%	$100-20=80$	$960 \div 80=12$	$12 \times 100=1200$	Rs 1200
SP=237.50, Loss=5%	$100-5=95$	$237.5 \div 95=2.5$	$2.5 \times 100=250$	Rs 250
SP=750, Loss=25%	$100-25=75$	$750 \div 75=10$	$10 \times 100=1000$	Rs 1000

Quick Application:

SP	Loss %	CP
540	10%	600
640	20%	800
475	5%	500
850	15%	1000

8. Applying Discount Formulas

$$\text{Discount} = \text{MP} \times (\text{Discount\%} \div 100)$$

$$\text{SP} = \text{MP} - \text{Discount}$$

$$\text{SP} = \text{MP} \times (100 - \text{Discount\%}) \div 100$$

Problem	Step 1: Find Discount	Step 2: Find SP	Answer
MP=1000, Disc=10%	$1000 \times 10\% = 100$	$1000 - 100 = 900$	Rs 900
MP=1500, Disc=20%	$1500 \times 20\% = 300$	$1500 - 300 = 1200$	Rs 1200
MP=2500, Disc=8%	$2500 \times 8\% = 200$	$2500 - 200 = 2300$	Rs 2300
MP=500, Disc=15%	$500 \times 15\% = 75$	$500 - 75 = 425$	Rs 425
MP=2000, Disc=25%	$2000 \times 25\% = 500$	$2000 - 500 = 1500$	Rs 1500

Quick Application:

MP	Discount %	Discount Amount	SP
800	10%	80	720
1200	20%	240	960
600	5%	30	570
3000	15%	450	2550

9. Finding Discount Percentage

$$\Sigma \text{ Formula: Discount\%} = (\text{Discount} \div \text{MP}) \times 100\%$$

Problem	Step 1: Find Discount	Step 2: Apply Formula	Answer
MP=1000, SP=900	$1000 - 900 = 100$	$100/1000 \times 100$	10%
MP=1500, SP=1200	$1500 - 1200 = 300$	$300/1500 \times 100$	20%
MP=500, SP=400	$500 - 400 = 100$	$100/500 \times 100$	20%
MP=2500, SP=2000	$2500 - 2000 = 500$	$500/2500 \times 100$	20%
MP=800, SP=680	$800 - 680 = 120$	$120/800 \times 100$	15%

Quick Application:

MP	SP	Discount	Discount %
1000	850	150	15%
600	540	60	10%
2000	1600	400	20%
500	450	50	10%

10. Applying Commission Formulas

Commission = Value $\times$ (Commission% $\div$ 100)

Amount Received = Value – Commission

Amount Received = Value $\times$ (100 – Commission%) $\div$ 100

Problem	Step 1: Find Commission	Step 2: Find Amount Received	Answer
Value=1,000,000, Comm=5%	1,000,000 $\times$ 5%=50,000	1,000,000–50,000=950,000	Rs 950,000
Value=500,000, Comm=3%	500,000 $\times$ 3%=15,000	500,000–15,000=485,000	Rs 485,000
Value=300,000, Comm=8%	300,000 $\times$ 8%=24,000	300,000–24,000=276,000	Rs 276,000
Value=200,000, Comm=10%	200,000 $\times$ 10%=20,000	200,000–20,000=180,000	Rs 180,000
Value=50,000, Comm=2%	50,000 $\times$ 2%=1,000	50,000–1,000=49,000	Rs 49,000

Quick Application:

Value	Commission %	Commission	Amount Received
200,000	5%	10,000	190,000
600,000	3%	18,000	582,000
400,000	10%	40,000	360,000
100,000	2%	2,000	98,000

11. Applying Profit/Loss with Multiple Items

Method 1: Find Total CP and Total SP

Problem	Calculation	Answer
Buy 50 coconuts at Rs 45 each, sell at Rs 60 each	CP=45 $\times$ 50=2250, SP=60 $\times$ 50=3000	Profit=750
Buy 100 mangoes at Rs 20 each, sell at Rs 18 each	CP=20 $\times$ 100=2000, SP=18 $\times$ 100=1800	Loss=200

Quick Application:

Items	Buying Price	Selling Price	Total CP	Total SP	Profit/Loss
50 at Rs 30	Rs 30	Rs 40	1500	2000	Profit 500
100 at Rs 15	Rs 15	Rs 12	1500	1200	Loss 300
200 at Rs 25	Rs 25	Rs 30	5000	6000	Profit 1000

Method 2: Find Profit/Loss per Unit, Then Multiply

Problem	Step 1: Per Unit	Step 2: Multiply	Answer
Buy 50 coconuts at Rs 45, sell at Rs 60	Profit per unit=60–45=15	15 $\times$ 50=750	Profit=750
Buy 100 mangoes at Rs 20, sell at Rs 18	Loss per unit=20–18=2	2 $\times$ 100=200	Loss=200

Quick Application:

Items	Unit CP	Unit SP	Per Unit P/L	Total P/L
80 at Rs 25	25	30	Profit 5	Profit 400
150 at Rs 40	40	35	Loss 5	Loss 750
60 at Rs 50	50	65	Profit 15	Profit 900

12. Applying Word Problems

Type 1: Mixed Buying and Selling

Problem	Steps	Answer
Vendor buys 60 kg at Rs 50/kg. Sells 20 kg at Rs 70, 15 kg at Rs 60, 10 kg at Rs 40, discards 15 kg.	CP=60×50=3000. SP=20×70=1400, 15×60=900, 10×40=400. Total SP=2700. Loss=300.	Loss=Rs 300

Quick Application:

Problem	Total CP	Total SP	P/L
Buy 100 kg at Rs 60/kg. Sell 40 kg at Rs 80, 30 kg at Rs 70, 30 kg at Rs 50	6000	3200+2100+1500=6800	Profit 800
Buy 200 at Rs 30 each. Discard 20, sell remaining at Rs 40	6000	180×40=7200	Profit 1200

Type 2: Profit/Loss with % and Discount

Problem	Steps	Answer
MP=Rs 2500, Discount=8%, CP=Rs 2000	Discount=200, SP=2300, Profit=300, Profit%=300/2000×100=15%	Profit=15%
CP=Rs 8000, Markup=25%, Discount=10%	MP=10000, Discount=1000, SP=9000, Profit=1000, Profit%=1000/8000×100=12.5%	Profit=12.5%

13. Applying Percentage of an Amount

Σ Formula: Amount × (Percentage ÷ 100)

Problem	Calculation	Answer
Find 25% of 480	$480 \times 25\% = 480 \times 0.25$	120
Find 35% of Rs 2500	$2500 \times 35\% = 2500 \times 0.35$	Rs 875
Find 8% of 750	$750 \times 8\% = 750 \times 0.08$	60
Find 15% of 1200	$1200 \times 15\% = 1200 \times 0.15$	180

Quick Application:

Amount	Percentage	Answer
500	20%	100
800	25%	200
1500	10%	150
2000	30%	600
100	5%	5

14. Application Checklist

Step 1: Identify what is given

■ Cost Price (CP)?
■ Selling Price (SP)?
■ Marked Price (MP)?
■ Profit/Loss percentage?
■ Discount percentage?
■ Commission percentage?

Step 2: Identify what to find

■ Profit or Loss amount?
■ Profit or Loss percentage?
■ SP or CP?
■ Discount amount or percentage?
■ Commission amount or percentage?

Step 3: Choose the correct formula

■ Profit = SP – CP
■ Loss = CP – SP
■ Profit% = (Profit ÷ CP) × 100%
■ Loss% = (Loss ÷ CP) × 100%
■ SP with profit = $CP \times (100 + P\%) \div 100$
■ SP with loss = $CP \times (100 - L\%) \div 100$
■ CP with profit = $SP \times 100 \div (100 + P\%)$
■ CP with loss = $SP \times 100 \div (100 - L\%)$

Step 4: Calculate carefully | **Step 5: Check your answer is reasonable**

15. Self-Test — Application Questions

No.	Question	Answer
1	CP=600, SP=750. Find Profit%	25%
2	CP=800, SP=680. Find Loss%	15%
3	CP=500, Profit=20%. Find SP	600
4	CP=1000, Loss=15%. Find SP	850
5	SP=660, Profit=10%. Find CP	600
6	SP=720, Loss=10%. Find CP	800
7	MP=1500, Discount=20%. Find SP	1200
8	MP=800, SP=680. Find Discount%	15%

No.	Question	Answer
9	Value=300,000, Commission=8%. Find Commission	24,000
10	Value=500,000, Commission=25,000. Find Commission%	5%

Revision Tip: Practice applying formulas to different situations. Start with simple problems, then move to word problems with multiple steps. Always check your answer is reasonable!